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Clinical Agreement in Wearable Technologies: A Bayesian Approach

Grado en Ciencia de Datos, 3ºB2

Escuela Técnica Superior de Ingeniería Informática

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Year: 2025-2026

Project: PROJIII-18.AGREEMENT

Abstract

This project evaluates whether wearable technologies can safely replace medical "Gold Standards" by assessing clinical agreement beyond simple correlation. Using a Bayesian hierarchical framework (**brms** and **INLA**), the study analyzed respiratory rate in COPD patients and heart rate across diverse skin tones. Findings reveal that while certain devices like the Apple Watch show superior resilience, many sensors remain sensitive to motion and physiological diversity, often failing Bayesian thresholds for clinical interchangeability. Specifically, devices such as chest-bands proved reliable only during rest, highlighting that validation must be context-dependent. This work provides a robust methodology and a deployment mockup to translate posterior distributions into actionable clinical verdicts.

Keywords: Measurement Agreement, Bayesian Inference, Clinical Validation, COPD, Heart Rate.

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1. Introduction

In modern clinical practice, portable and wearable devices are increasingly being used instead of hospital-based equipment. This change is often supported by showing that these devices have a high correlation with an established reference standard. However, correlation alone is not enough to prove that one device can replace another in clinical care [4]. A device may follow the same pattern as the reference method but still give readings that are consistently too high or too low. Correlation does not reveal this type of bias, and in clinical settings, it can lead to unreliable diagnostic decisions. For this reason, the key question is not whether two devices move in the same direction, but whether their measurements are close enough for one device to safely replace the other.

It should be clear from the start that this work is a descriptive agreement study, not a predictive one. The aim is not to predict the value of a signal, but to describe the difference between a portable device and its reference method, and to assess what that difference could mean clinically. Following the agreement framework proposed in Parker et al. [6], this is assessed using measures such as bias, limits of agreement (LoA), Coverage Probability (CP), Total Deviation Index (TDI), and Concordance Correlation Coefficient (CCC).

Two clinical areas are analyzed, each with a defined reference standard. In the University of Edinburgh COPD respiratory rate dataset Parker et al. [6], the reference standard is RRox, a respiratory rate measure derived from pulse oximetry and used as a clinical reference for patients with chronic respiratory diseases. In the heart rate dataset, the reference standard is the Bittium Faros 180, a medical-grade ECG patch recorded at around (~ 1000 Hz) and worn continuously throughout the protocol. Both datasets are described in more detail below. They are intentionally complementary: the COPD dataset is small, high-risk, and noisy, with limited structured data, while the heart rate dataset is larger and more diverse. This allows additional sources of variation to be examined, including activity type and skin tone based on the Fitzpatrick scale.

The contribution of this work is to extend the frequentist analysis Parker et al. [6] into a Bayesian framework. While a frequentist approach provides point estimates and confidence intervals, a Bayesian model gives full posterior distributions [5]. This makes it possible to express clinical decisions in terms of probabilities. For example, a device may have an estimated Coverage Probability (CP) above 80%, but it may still be unsafe to use if the posterior probability of meeting that threshold is low. By modelling observations as nested within patients and activities, the analysis also accounts for both within-subject and between-subject variability. This helps avoid underestimating uncertainty, which can happen when all observations are simply pooled together.

2. Methods

This section describes the data sources, the preprocessing pipeline and the modelling. The two datasets are presented at the same time because, although they are different in size and context, they share a common analytical determination: a unified representation centred on the paired difference, which feeds a hierarchical Bayesian agreement model. All preprocessing was implemented in R and the models were fitted with *brms* and *INLA*. The deployment prototype was built in Python with *bayesflow* and *Streamlit*.

The unit of analysis is the paired difference:

$$\text{diff} = \text{Device Value} - \text{Reference Value} \quad (2.1)$$

where zero means perfect agreement, and any consistent difference shows the device’s pattern of error.

Following the SMART framework from the seminars, the descriptive goal is defined through a clear *Measurement of Success*. A device is recommended for clinical use only if its Coverage Probability (CP)—which is the proportion of paired differences that fall within a clinically acceptable margin—meets the Bayesian decision rule: $P(\text{CP} \geq 80\%) \geq 70\%$. The acceptable margin is ± 6 rpm for respiratory rate and ± 5 bpm for heart rate.

The quality of the description is then evaluated using three criteria: (i) the posterior estimates of bias, LoA, CP, TDI and CCC and the width of their credible intervals; (ii) the satisfaction of the decision rule above, both globally and grouped by activity and skin tone; and (iii) the consistency of the final conclusion across datasets and alternative model specifications. Under this framework, the project does not only ask whether each wearable can replace its *Gold Standard*¹ in general, but also whether the wearable can be considered interchangeable under specific conditions.

2.1. Respiratory Rate in COPD Patients

The first dataset, released along Parker et al. [6] by the University of Edinburgh, has 385 observations from 21 patients with chronic obstructive pulmonary disease. Each row records the respiratory rate (RR) measured simultaneously by six wearable channels: two cameras (*cm1*, *cm2*), two PPG sensors (*ppg1*, *ppg2*), an impedance band (*imp*), an accelerometer (*acc*) and a chest band (*cb*), together with the clinical reference *RRox*, derived from pulse oximetry and adopted as the Gold Standard throughout the study (Table 2.1). Patients perform a fixed sequence of activities (rest, talking, walking, recovery), so observations are nested within both patient and activity, a hierarchy that any valid agreement model has to respect.

¹In clinical research, a Gold Standard is the benchmark measurement regarded as the most accurate and reliable for a specific condition, against which new methods are compared.

Variable	Description	Type
Id/Time/Obs	Identification, time and observation number	Categorical /Time
Activity	Type of activity performed	Categorical
RRox	Gold Standard	Numerical
RR	Rate measured with a camera (cm1/2), PPG (ppg1/2), impedance (imp), accelerometer (acc) and chest band (cb)	Numerical

Table 2.1: COPD Variables.

The exploratory analysis (Figures A.1 and A.2) shows two structural problems that end up shaping the rest of the pipeline. First, the marginal distributions of RR are different between channels: RRppg1 is shifted upwards, RRcm1 is shifted downwards, and RRimp shows quartiles close to RRox but with tails that suggest instability. Second, although the RRox relationship is roughly linear, missingness is high 52% and it is clearly non random: certain devices simply cannot measure during certain activities. This is exactly the kind of situation where correlation becomes misleading, since its value depends on the subset that happens to be observed, and it is what motivated the agreement approach used here in the first place.

Preprocessing was therefore conservative. Identifier and activity variables were converted to factors so the hierarchical models could group them correctly. Non physiological readings with zero or negative values were recoded as missing. Instead of dropping variables, only the device–activity pairs with more than 20% missing data were excluded, which preserves as much clinical signal as possible while keeping biased estimates from observed combinations out of the analysis. Outliers within the $1.5 \cdot \text{IQR}$ range were kept on purpose: in chronic respiratory patients, sudden spikes during exertion are physiological events, not measurement noise, and throwing them out would artificially inflate the apparent agreement of the devices.

2.2. Heart Rate in Wearable Devices

The second dataset ([3]) is used as an external validation of the framework on a much larger and more diverse population. It contains 238,595 heart rate records from 53 participants (with 32 F and 21 M, aged 18–54), distributed evenly across the six categories of the Fitzpatrick skin-tone scale, which is a dermatological classification that ranks skin types by their response to ultraviolet (UV) exposure, from Type I (very light, burns easily) to Type VI (very dark, rarely burns). This balanced design was on purpose: photoplethysmographic (PPG) sensors estimate heart rate from the optical absorption of green light by haemoglobin, and higher melanin levels attenuate that signal, so any agreement analysis of smart watches devices has to treat skin tone as a real source of variation rather than a nuisance covariate. Each participant followed a one-hour protocol divided into three rounds (Empatica E4 + Apple Watch 4; Fitbit Charge 2; Garmin Vivosmart 3 + Xiaomi Miband 3 + Biovotion Everion), with the Bittium Faros 180 ECG patch (1000 Hz) used as the Gold Standard across all rounds.

Each round followed the same activity sequence: seated rest for 4 min, paced deep breathing for 1 min, walking up to 50% HR max for 5 min, recovery for 2 min, typing for 1 min, so the design exposes every device to a comparable spectrum of motion artefacts. Variables are summarised in Table 2.2 and the full device specifications are reported in Table A.1.

Variable	Description (bpm)	Frequency/Variable
Faros 180ECG	Gold Standard	~1000 Hz
Watches	Apple Watch, Garmin, Fitbit, Miband	Variable
Pro Sensors	Empatica E4, Biovotion Ev-erion	~1 Hz
ID/SkinTone	ID ($n = 53$) and Fitzpatrick skin tone scale (1-6)	Categorical
Activity	Rest, Walking, Deep Breathing, Typing	Categorical

Table 2.2: Heart Rate Variables.

The exploratory analysis (Figures A.3 to A.7) tells a consistent story. The ECG is active throughout the protocol, while the smartwatches register less than 25% of the time (Figure A.3). The BPM range goes from a reasonable 40 BPM (athletic baseline) to an unreasonable 230 BPM, with all the extreme outliers concentrated in the ECG channel (Figure A.4). The chronological inspection of the protocol confirms that participants follow the prescribed Rest–Breathe–Activity–Type sequence (Figure A.5), but the volume of records is noticeably lower during rest and typing than during walking (Figure A.6) and the participant device coverage matrix is uneven, with Biovotion, Garmin and Xiaomi missing for many participants (Figure A.7).

This is the reason why the cleaning process had to deal with two different sources of noise: structural fragmentation (loss of skin contact, uneven coverage) and physiological improbability (ECG outliers). Rows with missing ECG were removed and incomplete device–reference pairs discarded, which reduced the dataset from 221,475 to 204,121 valid rows. Additionally, a physiological filter (30 to 200 BPM) was applied to the ECG channel to erase 91 residual artefacts, following the target heart rate guidelines established by the [1]. Since the heart rate dataset is two orders of magnitude larger than COPD, this stricter cleaning is affordable and provides credible intervals for the agreement metrics (CP, TDI, CCC) without sacrificing power.

2.3. Mineable View

After cleaning the datasets, both tables were transformed from a wide layout (one column per device) into a long format with one row per (subject, activity, device) measurement. The two datasets share the same analytical determination (one row, one paired comparison, one diff), but each retains the original column names produced by the pipeline. Table A.2 and Table A.3 document the schema of each mineable view and the origin of every field. The design follows the Bland–Altman rationale extended

by Parker et al. [6], so the comparison between the Bayesian results and the published frequentist baseline is direct.

2.4. Model Building

All models share the same logic: the difference $diff_{ij}$ is modelled as a Normal distribution around a mean that is decomposed into a global bias α , a patient-level random effect $u_i \sim N(0, \sigma_{\text{patient}})$ capturing physiological and anatomical variability, and when applicable, an activity-level random effect $v_j \sim N(0, \sigma_{\text{activity}})$ absorbing the shift in error caused by motion. Treating the 385 COPD rows or the 204,121 heart-rate rows as independent would lead to overconfident inferences; the hierarchy propagates within and between-subject variability into every posterior quantity, including the final clinical decision. A device is recommended only if $P(CP \geq 80\%) \geq 70\%$, with the clinically acceptable margin set to ± 6 rpm for COPD and ± 5 BPM for heart rate.

2.4.1 Model Building COPD

Two specifications were fitted (Table 2.3). The *Global model* is fitted separately for each device against RRox and includes both random effects, providing one verdict per device. The *Chest-band model* is fitted one activity at a time and includes only the patient effect, isolating the impact of motion on the best-performing device identified globally.

Model	Specification	Description
Global	$diff_{ij} \sim \text{Normal}(\mu_{ij}, \sigma^2)$ $\mu_{ij} = \alpha + u_i + v_j$ $u_i \sim \text{Normal}(0, \sigma_{\text{patient}})$ $v_j \sim \text{Normal}(0, \sigma_{\text{activity}})$	Fitted separately per device against RRox. The outcome is $diff = \text{device} - \text{RRox}$.
Chest-band	$diff_{ij} \sim \text{Normal}(\mu_{ij}, \sigma)$ $\mu_{ij} = \alpha + u_i$	Activity-stratified (one model per activity) to assess within-activity agreement.

Table 2.3: Global and Chest-Band Models.

2.4.2 Model Building HEART RATE

Two complementary frameworks were used (Table 2.4). The *INLA stratified model* fits one model per (activity $\times$ skin-tone) subgroup, exploiting the speed of Integrated Nested Laplace Approximation to detect changes in bias driven by motion artefacts and melanin interference. The *brms global model* mirrors the COPD global specification on the full dataset, decomposing total error into patient and activity variance components and producing a single hierarchical verdict per device.

Model	Structure	Description
INLA	$\text{diff}_{ij} = \alpha + u_i$	Stratified: One model per subgroup to identify changes in bias based on Activity and Skin Tone.
brms	$\text{diff}_{ij} = \alpha + u_i + v_j + \epsilon$	Global: Hierarchical model that decomposes total error into patient and activity variance.

Table 2.4: INLA and BRMS Models.

2.4.3 Model Building Bayesflow

To address the computational limitations of traditional MCMC and create an accessible tool for real-time clinical assessment, an **Amortized Bayesian Inference** framework was implemented using the *BayesFlow* library. Unlike models trained with INLA or *brms*, which require costly sampling via MCMC for each new dataset, a *BayesFlow* model trains a neural network that offers near-instant Bayesian inference for new data [7].

This methodology is executed for both datasets, creating two separate models, each specialized for the unique characteristics of the clinical datasets and their respective measurements. The evaluation of the *BayesFlow* model focuses on two main aspects: the stability of the amortized learning process and the precision of the resulting inference regarding key clinical parameters.

The learning process was measured using the Negative Log-Likelihood (NLL) loss function across 100 epochs. Both models exhibited a sharp initial drop followed by a smooth convergence towards an asymptote, indicating that the models effectively learned the posterior distribution of the data metrics.

Regarding the precision of the inferred clinical parameters, the Parameter Recovery plots (see Figure ?? and Figure ?? in the Appendix) demonstrate very high precision for the bias and residual error parameters, while showing moderate precision regarding patient variability.

In conclusion, these results validate the use of *BayesFlow* as a robust inference engine. It provides agreement metrics with minimal estimation error and a processing speed that meets the requirements for live clinical decision support.

2.4.4 Model Evaluation

The modelling process was developed iteratively rather than as a single fixed analysis. We first used frequentist agreement metrics as a baseline, following Parker et al. [6], and then moved to Bayesian hierarchical models to include uncertainty, patient-level variability and activity-level variability. For the COPD dataset, the first version was a global model fitted separately for each device. After seeing that the global verdict was hiding important differences between activities, we added an activity-stratified model for the chest-band, which was the best-performing device globally.

For the heart-rate dataset, the same logic was extended with two complementary approaches. INLA was used to fit stratified models by activity and skin tone, because it allowed faster subgroup analysis on a larger dataset. A global brms model was also fitted to keep a comparable Bayesian hierarchical baseline across devices. Across all iterations, the evaluation protocol remained the same: bias, limits of agreement, Coverage Probability, TDI, CCC and CIA were used, and the final clinical decision was based on the posterior rule $P(CP \geq 80\%) \geq 70\%$.

Since this is a descriptive study rather than predictive, classical hyperparameter optimization was not the main objective. Instead, the improvement focused on model structure, stratification variables, Bayesian uncertainty and clinical decision rules. The repeated calculation of agreement metrics across devices and subgroups allowed the analysis to become more systematic and made the final interpretation more context-aware.

To ensure good clinical recommendations, the parameters were set using meaningful margins. For heart rate, a margin of ± 5 bpm was used, which is considered an industry benchmark for clinical accuracy according to the ANSI/AAMI EC13 standard [2]. For the COPD dataset, the margin is ± 6 rpm, as it allows physiological peaks in chronic patients to be detected while limiting the impact of sensor noise.

To measure success, a Coverage Probability (CP) of at least 80% was necessary, ensuring that the vast majority of readings fall within the safe ranges. However, because point estimates can be misleading in noisy or small datasets, the Bayesian decision rule requires at least a 70% posterior probability that the device meets the threshold $P(CP \geq 80\%) \geq 70\%$. This requirement helps reduce overconfidence, ensuring that a device is recommended only when there is high certainty about the reliability.

3. Results

This section reports the agreement metrics from the four models in Section 2.4 and turns them into a clinical verdict using the rule $P(CP \geq 80\%) \geq 70\%$. Frequentist and Bayesian estimates are reported next to each other on purpose, because the gap between the two is the most interesting thing the analysis showed. The full posterior tables (means, 95% credible intervals, posterior probabilities and variance components) are in Appendix A, and they are referenced inline.

3.1. Bayesian Metrics

The six metrics used (Table 3.1) are the same ones suggested by [6], and each one was looked at twice: first as a frequentist point estimate, used as a baseline, and then as a posterior distribution from which a 95% credible interval and the posterior probability of meeting the clinical threshold were extracted. The recommendation is taken from that posterior probability, something that simply cannot be computed in the frequentist pipeline, which is the whole reason for moving to a Bayesian framework in the first place.

Detailed model results for the Global COPD, Stratified COPD, and Heart Rate models (INLA and BRMS) are provided in Table A.4 through Table A.11 within Appendix A.

Metric	Type	Definition & Clinical Threshold
Bias (α)	Systematic	Mean individual difference. A value near 0 indicates no systematic error.
Limits of Agreement (LoA)	Variability	The [95%] range of individual differences. Wide LoA signifies poor reliability.
Coverage Probability (CP)	Agreement	The % of errors within ± 6 rpm (COPD) or ± 5 BPM (HR). Threshold: $\geq 80\%$.
TDI	Risk	The maximum error expected in 95% of cases.
CCC	Precision	Combined measure of precision and accuracy (Threshold > 0.80).
CIA	Precision	Measures if the device adds more noise than the human variability itself.

Table 3.1: Frequentist decision metrics.

The Bayesian models also include a variance decomposition that splits the residual error into three sources: σ_{patient} (biological variability between subjects), σ_{activity} (induced by motion variability between tasks) and σ_{residual} (sensor noise). This is what

later allows the verdict to be conditional on activity and skin tone, instead of a single global number that hides everything underneath.

3.2. COPD Results: Global Model and Chest Band Model

In the global COPD analysis, no device meets the Bayesian threshold (Table A.4 and A.5). The Chest-band gets the highest agreement of all the devices ($CCC = 0.717$, $TDI = 10.1$ rpm, Table A.5), which means that, even for the best device, in 95% of measurements the error is expected to be up to 10 rpm away from the oximetry reading; this is almost twice the ± 6 rpm clinical margin. The accelerometer comes second ($CCC = 0.596$, Table A.5), and the camera and PPG devices are basically unusable because the missing data is too much and not random (52%), so the credible intervals come out very wide (Table A.4). The camera model, on top of that, throws 88 divergent transitions, which is a clear sign that the MCMC is unstable because there is just not enough data. The CIA values for these devices go above 1 (Table A.5), meaning the sensor adds more noise than the variability between patients; at that point, the device is not really measuring anything useful, but merely adding error on top of the natural differences between people.

The Chest-band is the only device that a frequentist framework would have approved (Table A.4 and A.5). Its frequentist CP is 80.8%, just above the 80% threshold, which would mean that the device stays within ± 6 rpm of RRox in 8 out of 10 readings. But the Bayesian model puts the mean CP at 72.1% and gives only $P(CP \geq 80\%) = 1.5\%$ (Table A.4), which means there is only a 1.5% chance that the device truly meets the 80% safety bar, and a 98.5% chance that it does not. What is happening is simple: the frequentist point estimate is sitting in the upper tail of a posterior distribution that is mostly below the threshold, so the apparent "pass" is really an issue of ignoring the uncertainty in a small dataset with hierarchical structure. In practice, trusting the frequentist verdict here would have meant putting a device into clinical use with a 98% probability of failing the safety criterion.

The activity-stratified Chest-band model brings the answer that the global model was hiding (Table A.6 and A.7). During activities with low movement, the posterior CP goes up to almost 1 and the device is *RECOMMENDED*, meaning that under those conditions every reading falls inside the ± 6 rpm clinical margin. During exercise, the posterior CP basically falls to 0%, which is the opposite extreme: almost no reading stays inside the safe range when the patient is moving. The variance decomposition (Table A.7) makes the reason obvious: σ_{activity} dominates σ_{patient} , so the global average is being pulled down by motion errors rather than by differences between patients. Therefore, the accurate conclusion is not that the chest-band can replace RRox, but that the chest-band can replace RRox under low-movement conditions, and only there, which is exactly the kind of conditional recommendation that motivated this project.

3.3. SMARTWATCHES Results: INLA Model and BRMS Model

The heart-rate analysis was run with two complementary inference ways. The INLA stratified model fits one cell at a time (Table A.8 to A.10) and the brms global model gives one hierarchical verdict per device (Table A.11). Both end up agreeing on the same thing, that no device accomplishes the Bayesian threshold in clinical terms, none of the smartwatches tested can be safely used as a replacement for an ECG patch under the criteria adopted in this work.

The INLA results show a lot of heterogeneity. The Apple Watch is the most consistent of all, with CP between 0.35 and 0.45 in the Fitzpatrick scale (Table A.10) and only a little fluctuation in the activities (Table A.9), which means that only 35 to 45% of the readings stay in the ± 5 BPM margin of the ECG, depending on the skin tone of the person, this is far below the 80% bar. Biovotion and Empatica fall apart during everyday tasks, with CP values close to zero (Table A.8), meaning almost none of their readings are in the clinical range while the user is moving. Garmin shows an error of -119.64 BPM (Table A.8), which is too big to be real and looks more like a synchronisation problem in the export than an actual sensor failure, on average, Garmin reports a heart rate 120 BPM lower than the ECG, an error of that size can't come from the sensor and certainly reflects misaligned timestamps. The CP and CCC distributions broken down by Fitzpatrick type (Figures A.12 and A.13) confirm that motion noise and melanin interference act as non random error triggers, agreement gets worse as skin tone darkens, so the same device that works well on lighter skin (Type I and II) becomes unreliable on darker skin (Type V and VI), which is exactly what would be expected from the optical attenuation that PPG suffers with higher melanin levels.

The brms model (Table A.11) confirms the same under a single hierarchical specification. Even the best device, Apple Watch, ends up with a global posterior CP of 0.005 and a TDI = 39.85 BPM (Table A.11), meaning only 0.5% of readings are in the ± 5 BPM clinical range, and in 95% of cases the error can be up to 40 BPM, almost eight times the safe range. Garmin's CIA of 1 (Table A.11) again indicates a data issue rather than the sensor failing. The variance decomposition (Table A.11) returns $\sigma_{\text{activity}} > \sigma_{\text{patient}}$, the same pattern as COPD: that motion is the principal cause of disagreement, and the skin tone accentuates it.

3.4. Deployment Mockup

A prototype (Figures A.14 and A.15) has been developed using *Streamlit* to automate concordance analysis. This interface is designed for descriptive agreement analysis, not a predictive model. The user uploads a dataset, selects the Gold Standard column and the device column, chooses the device type only to apply the appropriate clinical tolerance, and receives agreement metrics with an interpretable clinical verdict instantly. This verdict translates the statistical results (like CP and TDI) into a simple, actionable phrase for the user.

4. Discussion

4.1. Beneficiaries, Limitations & Risks

The main value of the project is not just ranking the different devices, but providing a clinical decision framework that allows determining when a device can be interchangeable with its standard. With the COPD case, it can be seen that a device may not be suited globally, but still useful under certain circumstances. This distinction is valuable because it avoids rejecting a device that can still be useful in controlled conditions (like during resting activities), as well as avoiding accepting a device in situations where the device itself is unreliable (like activities with movement).

The main beneficiaries are medical experts and device developers. The medical experts could use this analysis to see whether a device is reliable enough for patient monitoring. If a device is cheaper or more comfortable to use than the gold standard, this type of analysis could facilitate clinical decision making. It can also prevent unsafe substitution by helping see in which conditions the device fails.

As for the developers, they could use the results to better understand where the measurement errors come from, like from different types of activities. This kind of information can help them identify the context where their performance is poor and focus on improving their designs there. It can also help to explain to the users when the device is not recommended.

The main limitation is the data quality; for example, in the COPD dataset there is a lot of missing data, which limits the stability of some estimates and makes some combinations of device and activity difficult to analyse. In the heart-rate dataset, its size is much larger, but the devices were not all worn simultaneously in the same round, adding variability.

A second limitation is that the data from both datasets come from a controlled experiment, so the results cannot be applied directly to all populations or real-world settings. The devices' performance may change, for example when the sensors are placed differently, or are used in patients that are not in the datasets.

A third limitation would be the thresholds. For example, in a higher-risk clinical setting, such as the identification of a chronic disease, stricter thresholds might be required. Therefore, the final recommendation depends on the clinical context in which it is applied.

The main risk is using the model verdict as if it were a clinical decision. What the model says should not replace medical judgement; it should be used as a support tool for decision making. A "recommended" result should not be interpreted as permission to use the device and should only be accepted after a further analysis.

Another risk is overgeneralising the results. A device that performs well in one subgroup or dataset should not be assumed to be reliable in every situation. For

example, the COPD results show that a device may work under low-movement conditions but fail during activities with stronger movement. Using the device outside the conditions where it was validated could lead to unsafe conclusions.

The novelty of this project lies not in the invention of new statistical theories but in the strategic transition from traditional frequentist point-estimates to a superior Bayesian hierarchical framework for clinical agreement. While industry standards often rely on fixed values for bias and precision that ignore the inherent uncertainty of repeated measures, this approach propagates variance through every level of the hierarchy to provide a more rigorous assessment of sensor reliability. By implementing a conditional decision rule, this work moves beyond simplistic verdicts (yes/no) toward a context-aware validation that accounts for specific patient activities and skin tones. This shift toward personalised safety metrics aligns with modern regulatory trends, such as those advocated by the **FDA** regarding the use of Bayesian statistics to better manage clinical risk and variability in medical device development [9].

4.2. Legacy, Dissemination & SDGs

The project can grow in several directions. The first improvement would be to validate the pipeline on more clinical datasets, especially data from real-world environments rather than controlled experiments. The second would be to improve the mockup by adding automatic data-quality checks, missing-data reports, and downloadable PDF summaries. Finally, testing the model on new participants, especially for the COPD dataset, would help confirm whether the agreement conclusions generalise to broader populations.

Many lessons were learnt during this project. The first is that high correlation does not imply agreement therefore, correlation alone is insufficient to decide whether a wearable device can replace a clinical reference. The second is that uncertainty must be included in the final decision, as a device may appear acceptable but remain unsafe if the posterior probability of meeting the threshold is low. Finally, building the webpage showed that a statistical model only has real value when its results are interpretable and actionable for the end-user. Another possible improvement would be to add preexisting suppositions based on clinical knowledge, so that the framework can include expert advice in a clear way and move the Bayesian decision rule even further from the standard frequentist results. This would be especially useful in small datasets like the COPD one, where prior information has the biggest impact on the final result.

In regards to Sustainable Development Goals, the project is mainly connected to SDG 3: Good Health and Well-being, addressing in particular target 3.4 (reducing premature mortality from non-communicable diseases such as COPD and cardiovascular conditions through safer continuous monitoring) and target 3.d (strengthening health-risk management capacity by giving public systems a transparent way to decide when low-cost wearables can be trusted as substitutes for hospital-grade equipment). The project also aligns with SDG 9: Industry, Innovation and Infrastructure, specifically target 9.4, since it upgrades the wearable-evaluation infrastructure both methodologically, replacing point estimates with full posterior decisions, and operationally, through a *Streamlit* prototype that lets manufacturers and clinicians re-evaluate new devices

without rebuilding the analytical pipeline from scratch [8].

For dissemination, there are two main options. The first one is to publish the *R* code, the cleaning scripts and the *Streamlit* prototype as open source on a public repository, so that other research groups working with the COPD and similar datasets can use and extend the analysis. The second one is to share the project as a short tutorial for biomedical engineering or biostatistics courses, using Parker et al. [6] as a starting point.

4.3. Use of technology

Throughout this project, the primary tool used was R Studio, due to its statistical packages. One of the packages used is tidyverse (dplyr, tidyr, ggplot2) for data visualisation and cleaning. Thanks to ggplot2, various box plots, matrices and other graphs were created throughout the report.

Additionally, the brms package was used, which is essential for creating Bayesian models. In addition, INLA (Integrated Nested Laplace Approximation) was used to analyse the Heart Rate dataset, as it contained a large volume of data and, in such cases, this package is more efficient than brms.

Finally, the report was done with overleaf and the mock-up was created as a web application using the Python library Streamlit to allow a good translation from the statistical model to a visual data product. This way a user can upload a dataset, select the gold standard and device columns and obtain an interpretable agreement verdict.

4.4. Seminar and AI use

During the elaboration of this project, several concepts from the seminars were used to structure the project, as well as supporting tools like artificial intelligence (AI).

First, the distinction between clinical/business goals and data analysis goals was used. The clinical goal was to decide whether a wearable device could replace a Gold Standard, while the data goal was to measure agreement between both methods. This helped define a clear Measurement of Success, where a device is recommended only if $P(CP \geq 80\%) \geq 70\%$, using ± 6 rpm for respiratory rate and ± 5 bpm for heart rate. This was appropriate because our project needed a clinical decision rule, not only good statistical results.

Moreover, the task taxonomy from unit 3 of the seminar classes was used to help determine the type of project to choose. The problem does not focus on predicting future values nor on classifying observations, but rather consists of determining whether a wearable device is clinically substitutable for a gold standard under certain circumstances, so the task is framed as descriptive, instead of predictive or classificatory.

This affected the choice of metrics, as the seminars recommended the use of adequate metrics. In this case, common regression metrics such as RMSE or R^2 were not suitable because, as stated by Parker et al. [6], correlation does not imply concordance.

Instead, more specific agreement metrics were selected: bias, LoA, CP, TDI, CCC and CIA.

The seminars on iterative improvement were also used. Firstly, global Bayesian hierarchical models were built, but then stratified analyses by activity and skin tone were added when seeing that the global results were hiding relevant differences. This was especially useful for the Chest-band, which was not globally substitutable but became recommendable under low-movement conditions.

Finally, AI tools such as *ChatGPT*, *Gemini* and *Perplexity* were used for text structuring and code assistance. Its main potential was to speed up the workflow by helping with section organisation and clarify technical explanations. However, all the statistical interpretations, metric selection and model conclusions were done by the team itself. Therefore, AI was more of a support tool, rather than a decision-making tool.

AI also had limitations. For example, in the COPD dataset, it suggested removing all rows with NA values, but this was not done because missing values were not random, as some devices failed more often in specific activities, which would have biased the agreement analysis. It also tended to simplify conclusions to one global verdict per device, while our stratified results showed that some devices, such as the Chest-band, were useful only in specific conditions. For this reason, all AI outputs were checked and adapted by the team.

5. Conclusions

This project shows that clinical agreement cannot be reduced to correlation or to a single yes/no result. Following the agreement framework proposed by Parker et al. [6], the analysis first evaluates bias, Limits of Agreement, Coverage Probability, TDI and CCC, but extends their interpretation through a Bayesian hierarchical approach. This changes the clinical reading of the COPD results, where the chest band appears acceptable under the frequentist criterion, with a CP of 80.8%, but the Bayesian model estimates a posterior CP of 72.1% and only a 1.5% probability of truly exceeding the 80% threshold. Therefore, it should not be considered globally substitutable for the Gold Standard.

However, the Bayesian model also avoids the opposite mistake of rejecting the device completely. The model with an activity analysis shows that the chest band can be reliable under low movement conditions such as standing, sitting, lying and treadmill activity, while its agreement deteriorates during more intense movement. The conclusion is therefore not simply that the device works or fails, but that its reliability depends strongly on the clinical context in which it is used.

The heart-rate dataset confirms the same idea in a larger and more diverse setting, but with a stricter outcome. The Apple Watch is the strongest device in the analysis, yet its Coverage Probability remains well below the 0.80 clinical threshold, around 0.35-0.45 across skin-tone groups. This means that being the best smartwatch in the dataset does not make it clinically interchangeable with an ECG patch.

The global model reinforces this conclusion, with a TDI close to 40 bpm for the Apple Watch, far beyond the accepted ± 5 bpm margin. Therefore, these smartwatch devices may be useful for day to day monitoring, but they should not be replacing ECG in a clinical setting. Nonetheless this analysis helps in ranking the different devices, with the Apple Watch performing significantly better than its competitors.

At the end of this analysis, the results show that current wearable devices, on their own, are not enough to meet the standards required for full clinical substitution. However, this limitation is also one of the most important findings of the project. Evaluating these tools is not only about giving a simple “yes” or “no” answer, but about identifying how far they can go and under which specific conditions they become useful.

Overall, the main contribution of this work is the framework itself, a Bayesian agreement pipeline that turns device evaluation into a probabilistic and context-aware decision. Instead of answering only whether a wearable works, the analysis answers for whom and under which conditions it can be trusted.

6. References

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A. Appendix

A.1. Exploratory Data Analysis

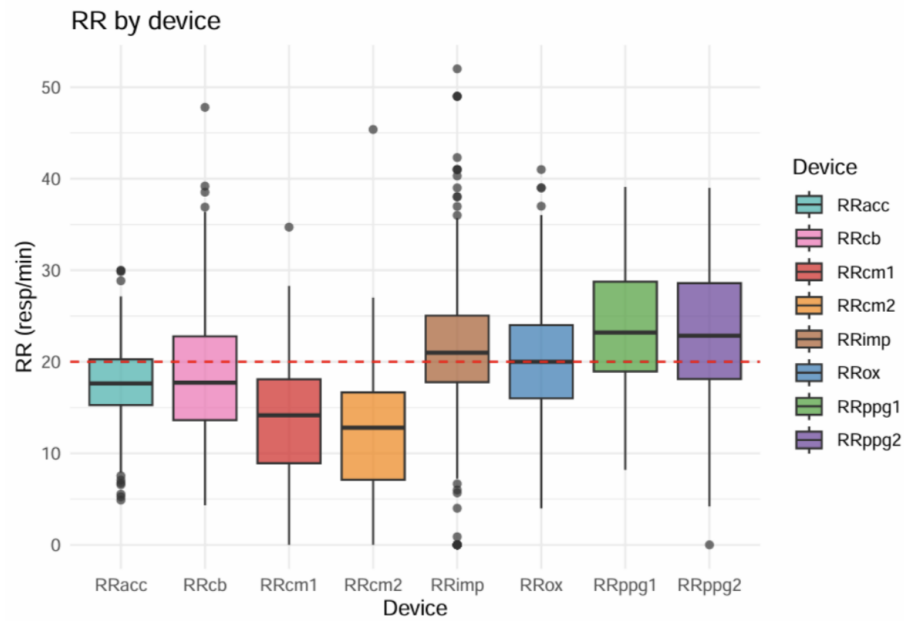


Figure A.1: Boxplots RR by Device

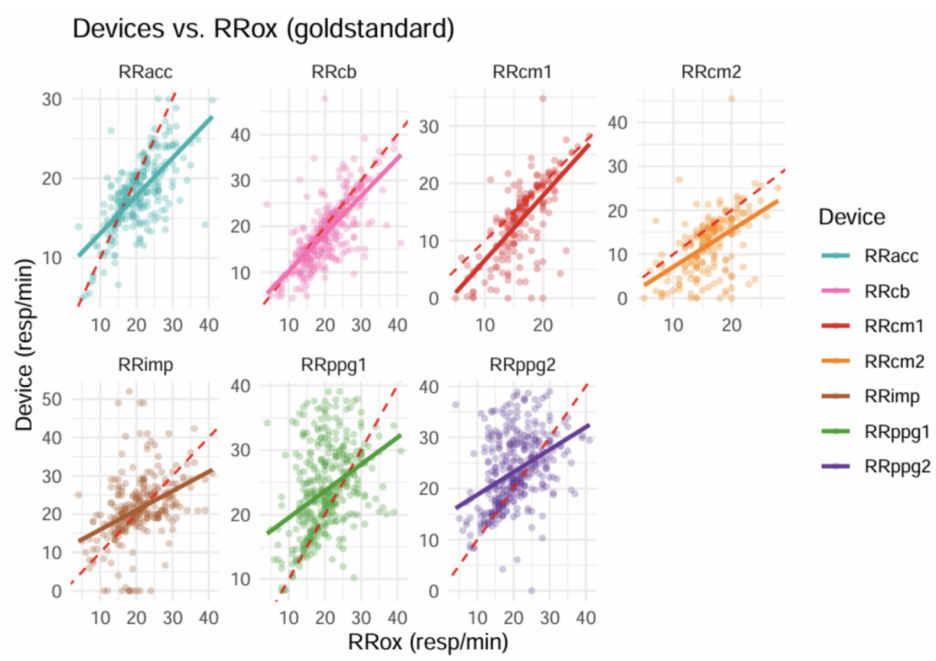


Figure A.2: Devices vs GoldStandard

Device	Manufacturer	Location	Release	Software
Apple Watch 4	Apple Inc.	Cupertino, CA	Fall 2018	5.1.3
Fitbit Charge 2	Fitbit, Inc.	San Francisco, CA	Fall 2016	22.55.2
Garmin Vivosmart 3	Garmin Ltd.	Olathe, KS	Spring 2017	5.10
Xiaomi Miband 3	Xiaomi Corp.	Beijing, China	Spring 2018	NA
Empatica E4	Empatica Inc.	Milano, Italy	NA	Summer 2019
Biovotion Everion	Biovotion AG	Zurich, Switzerland	NA	Summer 2019
Bittium Faros 180 (GS)	Bittium Corp.	Oulu, Finland	—	—

Table A.1: Device Specifications for the heart rate dataset

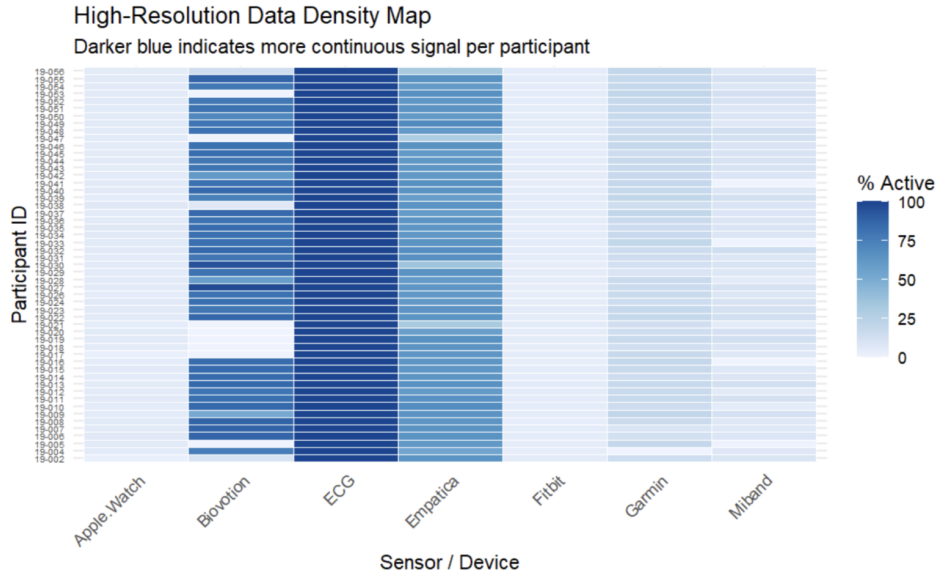


Figure A.3: Sensor Activity Profile per Participant

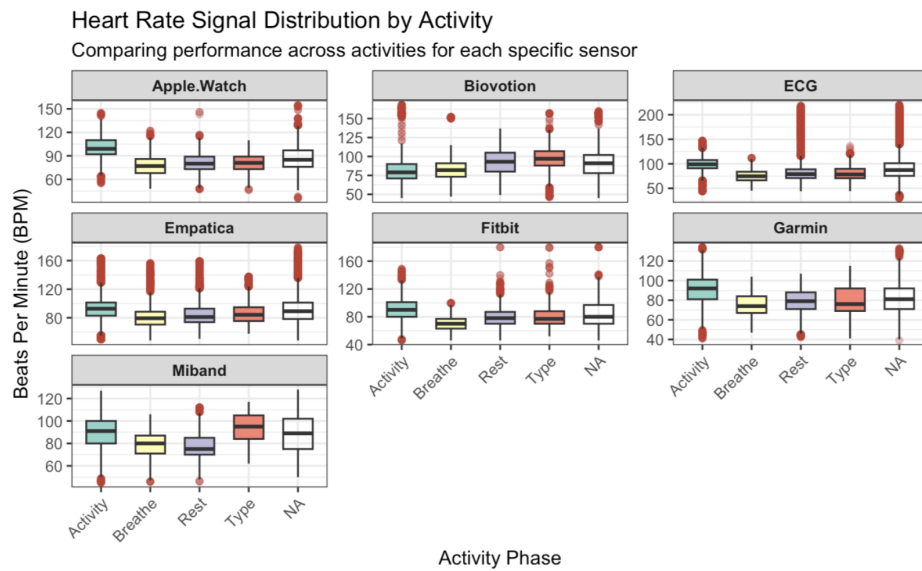


Figure A.4: Heart Rate Distribution by Activity and Device

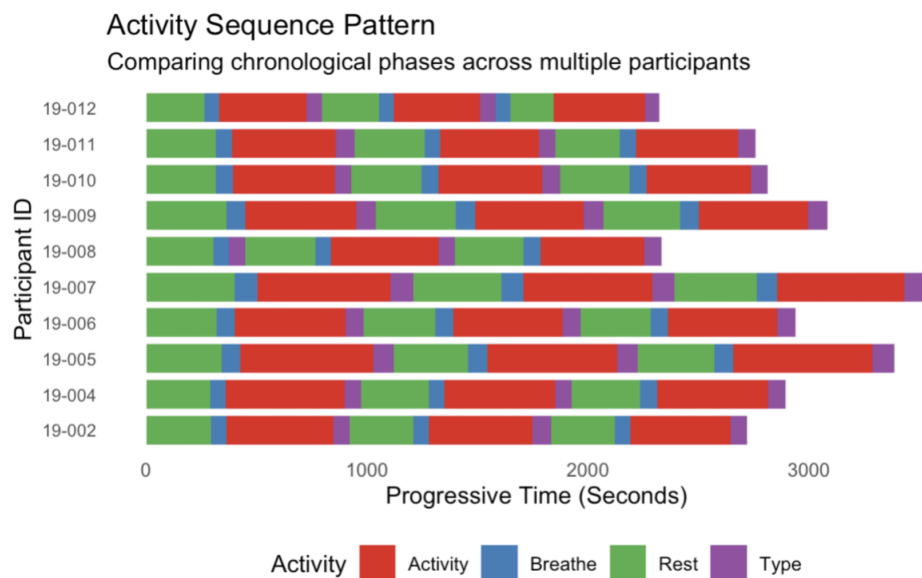


Figure A.5: Activity Sequence Pattern

Column	Type	Origin	Role
PatientID	Categorical (factor)	Original PatientID	Grouping variable for patient random effect u_i .
Obs	Integer	Original observation index	Preserves measurement order; not used as predictor.
Activity	Categorical (factor)	Original Activity column (rest, talking, walking, recovery)	Grouping variable for activity random effect v_j .
Device	Categorical (factor)	pivot_longer (names_to = "Device")	Identifies the wearable; one model fitted per device.
RR	Numerical (rpm)	pivot_longer (values_to = "RR")	Device measurement paired with RRox. ≤ 0 set to NA; device-activity pairs with $> 20\%$ missing excluded.
RRox	Numerical (rpm)	Original RRox	Gold Standard reference value.
diff	Numerical (rpm)	$RR - RRox$	Outcome variable of Bayesian models.

Table A.2: Mineable view for COPD dataset

Column	Type	Origin	Role
ID	Categorical (factor)	Original participant identifier (n = 53)	Grouping variable for patient random effect u_i .
Skin.Tone	Categorical (Fitzpatrick I–VI)	Original Skin.Tone column	Stratification variable; key for PPG analysis.
Activity	Categorical (factor)	Original Activity column (rest, deep breathing, walking, typing)	Grouping variable for activity random effect v_j .
Device	Categorical (factor)	pivot_longer (names.to = "Device")	Identifies which wearable produced each row; one model is fitted per device.
HR_Device	Numerical (bpm)	pivot_longer	Device measurement (bpm).
HR_GoldStandard	Numerical (bpm)	Original ECG	Reference value (30–200 BPM filter).
diff	Numerical (bpm)	$HR_{Dev} - HR_{GS}$	Outcome variable of Bayesian models.

Table A.3: Mineable view for HEART RATE dataset

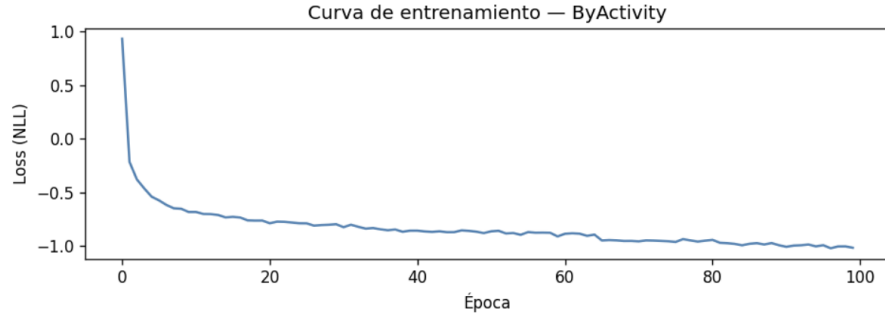


Figure A.8: COPD bayesflow Recovery Plot

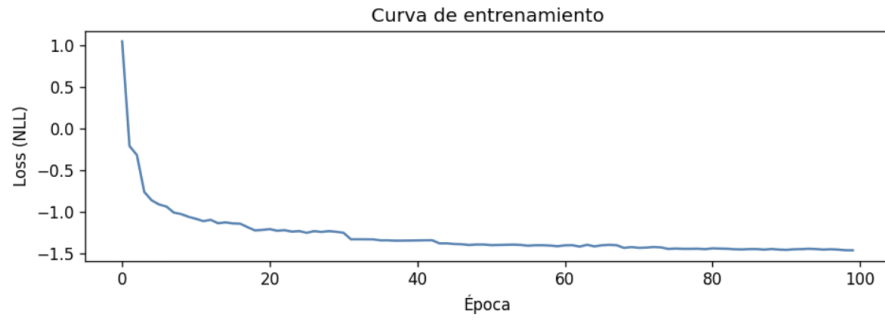


Figure A.9: COPD bayesflow Recovery Plot

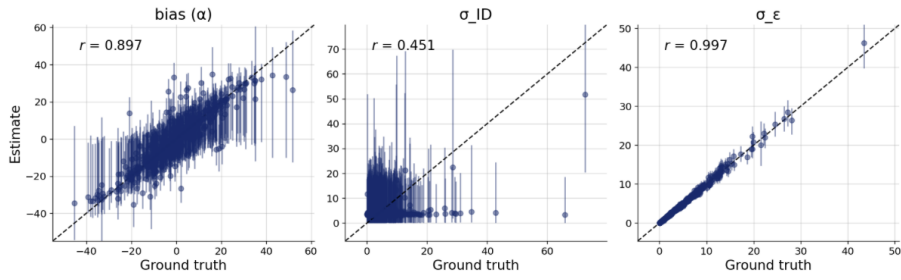


Figure A.10: Smartwatches bayesflow Recovery Plot

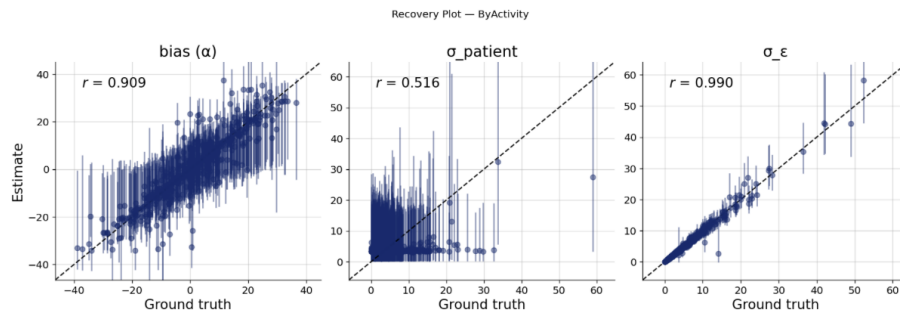


Figure A.11: S bayesflow Recovery Plot

A.2. COPD Agreement Results

Device	Bias (rpm)	LoA (rpm)	CP_freq	CP_bayes (95%)	P(CP $\geq$ 80%)
Camera 1	-3.29	[-14.3, 7.7]	77.2%	63.1% [38.8%, 75.7%]	0%
Camera 2	-5.83	[-21.1, 9.5]	69.0%	42.9% [20.2%, 60.7%]	0%
PPG 1	2.86	[-11.9, 17.6]	62.4%	53.9% [46.4%, 59.9%]	0%
PPG 2	2.54	[-12.7, 17.8]	63.2%	53.2% [45.7%, 59.0%]	0%
Impedance	-4.57	[-27.4, 18.2]	55.2%	36.4% [27.9%, 42.9%]	0%
Accelerometer	-3.46	[-13.9, 7.0]	82.2%	63.6% [42.7%, 78.6%]	1.3%
Chest-band	-1.27	[-11.7, 9.2]	80.8%	72.1% [60.3%, 79.4%]	1.5%

Table A.4: Global Bayesian Agreement: Bias, LoA and Coverage Probability

Device	TDI	CCC	CIA	Verdict
Camera 1	12.6	0.561	0.132	Not substitutable
Camera 2	18.9	0.363	-0.457	Not substitutable
PPG 1	15.2	0.338	0.19	Not substitutable
PPG 2	15.4	0.353	0.199	Not substitutable
Impedance	23.8	0.234	-0.927	Not substitutable
Accelerometer	12.3	0.596	0.678	Not substitutable
Chest-band	10.1	0.717	0.678	Not substitutable

Table A.5: Global Bayesian Agreement: TDI, CCC and Clinical Verdict

Activity	Bias(rpm)	LoA(rpm)	CP_Freq	CP_bayes	P(CP $\geq$ 80%)
Standing	-1.8	[-5.6, 2.0]	100%	98.1%	100%
T Flat	-0.85	[-3.8, 2.1]	100%	99.8%	100%
TS 4%	0.04	[-2.9, 3.0]	100%	99.9%	100%
Sitting	-1.43	[-7.7, 4.9]	89.7%	90.9%	99.8%
Lying	0.47	[-4.8, 5.8]	90.5%	96.2%	99.7%
Sweeping	1.72	[-8.4, 11.8]	66.7%	72.5%	15.9%
Fast 6mWT	1.34	[-12.5, 15.2]	81%	59.2%	0.2%
Slow 6mWT	2.14	[-11.9, 16.2]	81%	57.4%	0.1%
Climbing S	-6.57	[-20.5, 7.4]	52.4%	42.4%	0%
Lifting	-4.98	[-15.6, 5.6]	52.4%	55.0%	0%
Stnd&W	-3.82	[-14.0, 6.4]	62.7%	62.9%	0%

Table A.6: Activity-Stratified Agreement: Bias, LoA and CP

Activity	TDI	CCC	CIA	Verdict
Standing	4.96	0.863	0.855	RECOMMENDED
T Flat	3.29	0.965	0.964	RECOMMENDED
TS 4%	2.75	0.968	0.967	RECOMMENDED
Sitting	6.71	0.719	0.651	RECOMMENDED
Lying	5.07	0.74	0.749	RECOMMENDED
Sweeping	10.2	0.613	0.371	NO
Fast 6mWT	13.3	0.445	0.572	NO
Slow 6mWT	14	0.32	-0.19	NO
Climbing S	18.3	0.118	-0.432	NO
Lifting	13.9	0.381	0.106	NO
Stnd&W	12.4	0.474	0.24	NO

Table A.7: Activity-Stratified Agreement: TDI, CCC and Verdict

A.3. Heart Rate Agreement Analysis

Device	Bias (α)	TDI (95%)	CP (5 BPM)	CIA	CCC
Apple Watch	-0.17	18.52	0.405	0.05	0.85
Fitbit	-4.20	34.54	0.228	0.14	0.50
Miband	-2.44	37.43	0.209	0.23	0.35
Empatica	-0.19	38.68	0.198	0.22	0.34
Biovotion	-3.99	46.73	0.164	0.16	0.21
Garmin	-119.64	1000	0.000	1.00	0.59

Table A.8: Global Bayesian Agreement (INLA)

Device	Rest (CP)	Breathe (CP)	Activity (CP)	Type (CP)
Apple Watch	0.43	0.36	0.54	0.59
Fitbit	0.30	0.45	0.22	0.25
Miband	0.25	0.35	0.00	0.16
Garmin	0.00	0.47	0.00	0.33
Empatica	0.23	0.00	0.21	0.00
Biovotion	0.20	0.00	0.00	0.00

Table A.9: Global Bayesian Agreement (INLA) by Activity

Device	Tone 1	Tone 2	Tone 3	Tone 4	Tone 5	Tone 6
Apple Watch	0.42	0.41	0.35	0.45	0.41	0.40
Fitbit	0.00	0.27	0.22	0.20	0.23	0.21
Miband	0.23	0.24	0.21	0.18	0.21	0.23
Empatica	0.23	0.22	0.19	0.18	0.20	0.21
Garmin	0.00	0.00	0.27	0.21	0.00	0.27
Biovotion	0.20	0.19	0.18	0.18	0.16	0.13

Table A.10: CP by Skin Tone (Fitzpatrick Scale)

Device	CCC	LoA Inf	LoA Sup	TDI (95%)	CP (5 BPM)	CIA
Apple Watch	0.123	9.893	42.647	39.848	0.00575	0.160
Empatica	0.022	38.052	92.692	87.923	0.00013	0.120
Fitbit	0.061	21.094	73.902	70.053	0.00075	0.144
Biovotion	-0.005	87.639	131.245	128.278	0.00000	0.725
Garmin	0.053	21.915	61.652	58.175	0.00013	0.311
Miband	0.013	43.501	91.279	87.056	0.00000	0.231

Table A.11: Global Bayesian Agreement (brms)

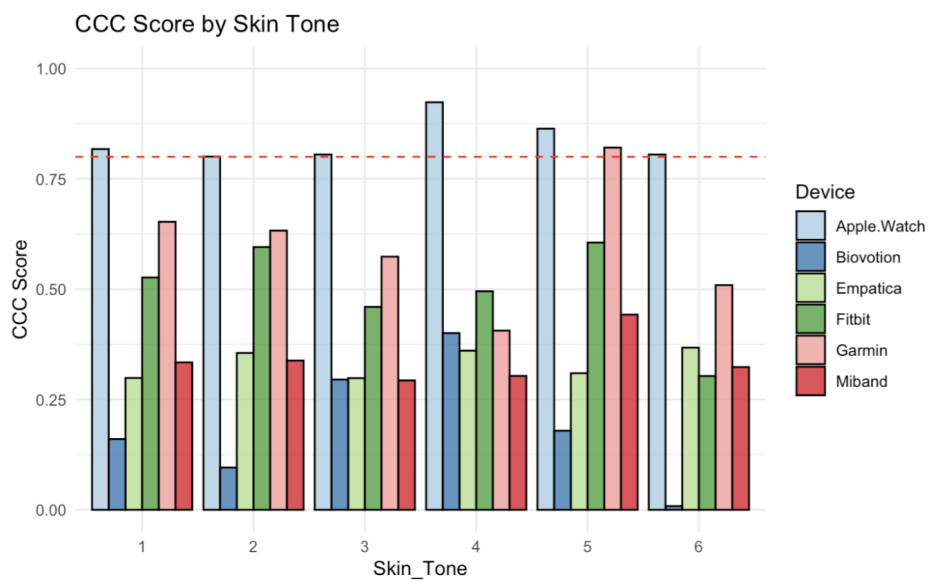


Figure A.12: CCC Score for device by Skin Tone

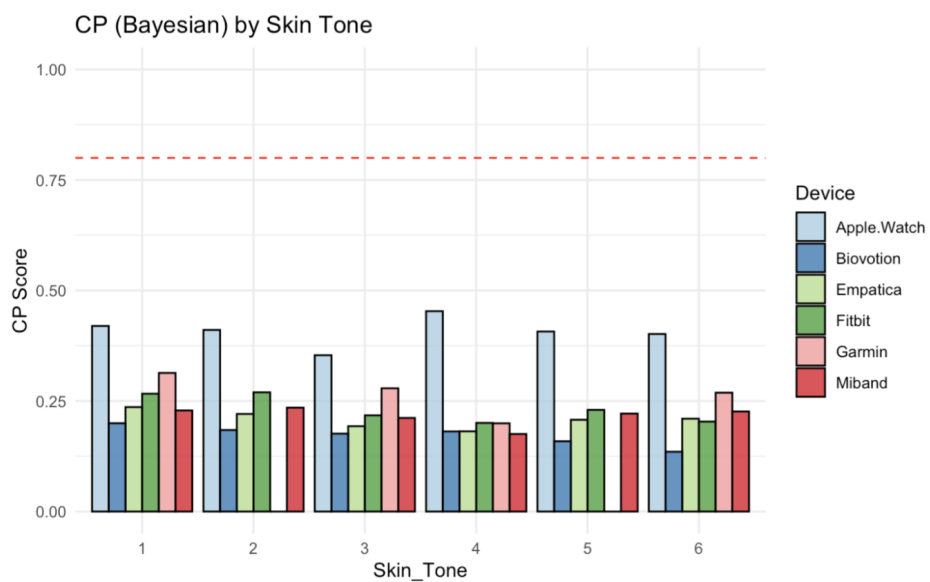


Figure A.13: CP Score for device by Skin Tone

A.4. Deployment Mockup

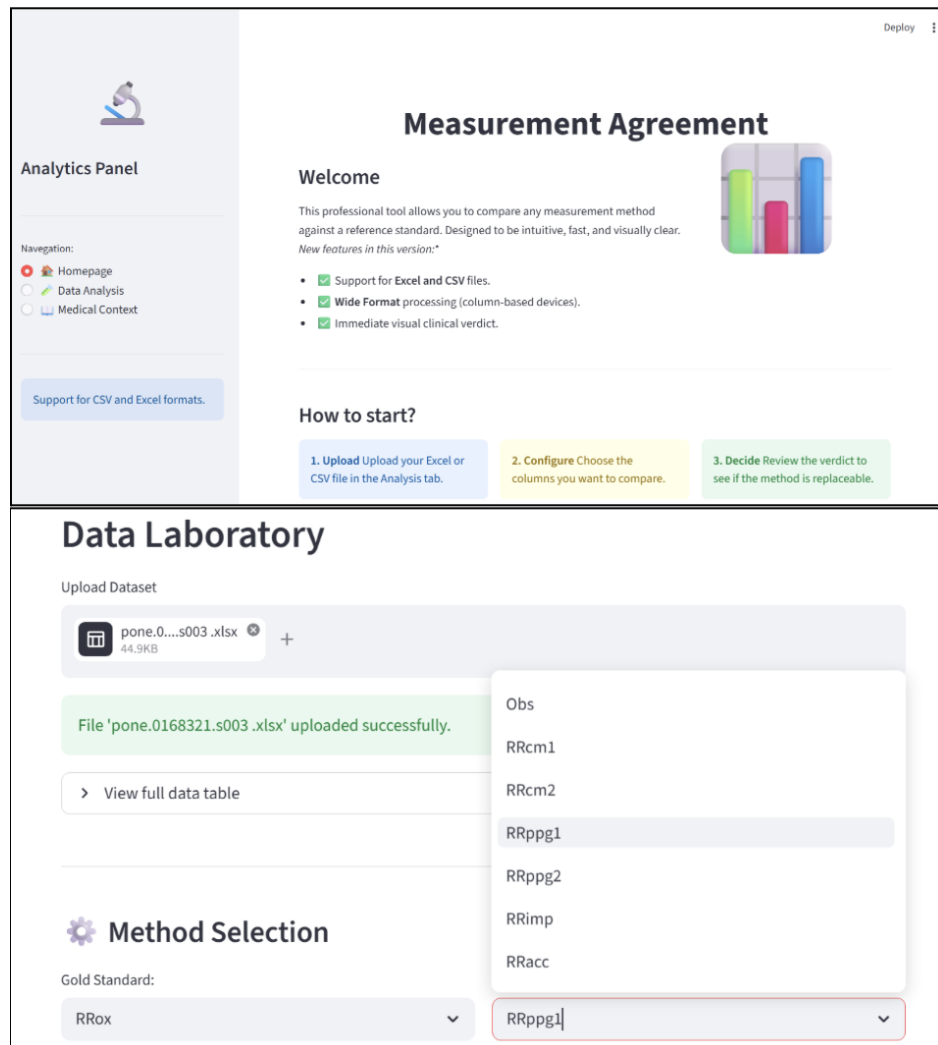


Figure A.14: Mockup Interface

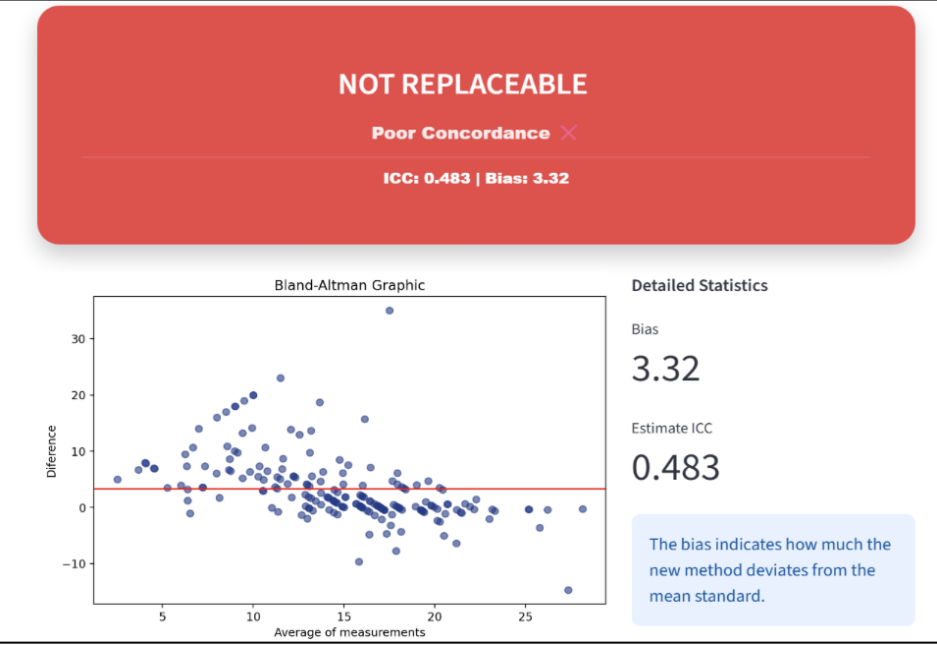


Figure A.15: Mockup Results